Zhenyu Zhang

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Summary

Applied machine learning researcher and software engineer with an M.Sc. in Computational Neuroscience and a background in computer engineering. Experienced in developing deep learning models for large-scale clinical EEG and MEG analysis, with expertise in representation learning and time-series modeling. Skilled in designing rigorous experiments, evaluating model performance, and translating research concepts into practical implementations. Brings additional industry experience in production C/C++ development, software testing, and CI/CD.

Experience

Simon Fraser University **Burnaby, BC**
Graduate Researcher *Aug 2024 – Present*

- Designed a reader-conditioned ordinal learning framework jointly modeling neurologist identity and diagnostic confidence from clinical EEG reports; evaluated on 35,946 routine recordings across multiple Fraser Health Authority hospital sites.
- Adapted and evaluated large EEG foundation model for clinical abnormality classification; Evaluated pretrained EEG foundation models and investigated representation dynamics using information-theoretic and temporal analyses.
- Implemented calibration analyses, ordinal coherence metrics to quantify uncertainty in clinical decision-making.
- Built machine learning pipelines for 40k+ clinical EEG recordings using PyTorch, HDF5, and HPC infrastructure.

Huawei Technologies **ShangHai, CN**
Software Engineer *Apr 2023 – Apr 2024*

- Built system-space C/C++ services spanning normal execution and TEE APIs, responsible for public-key validation, local token issuance/management, challenge routing, biometric invocation, and token expiry tracking.
- Authored 10k lines of production/support code across C/C++, QA tooling, and TypeScript, excluding mocks. Implemented HarmonyOS device-side authentication services for IFAA contactless payment and FIDO-based banking authentication

National Research Council Canada **Ottawa, ON**
Research Intern *Aug 2022 – Jan 2023*

- Investigated transformer-based foundation models for sound event detection and human activity recognition in elder home care settings.
- Developed self-distillation pipelines for weakly supervised learning under limited-label conditions; evaluated knowledge distillation strategies across self-supervised audio representations.
- Conducted large-scale experimental evaluations and statistical analyses of representation transfer learning across multiple model architectures.

McMaster Children's Hospital **Hamilton, ON**
Research Intern *Jan 2022 - Aug 2022*

- Developed computer vision tools for ultrasound-assisted colon motility tracking and virtual colonoscopy; validated image-based motility estimation against clinical ground truth.
- Built image processing and quantitative analysis pipelines for non-invasive gastrointestinal assessment using deep learning applied to colonoscopy imagery.

Education

Simon Fraser University **Burnaby, BC**
M.Sc. in Computational Neuroscience *2024 – 2026*

- Thesis: *Reader-Aware Modeling of Routine Clinical EEG*

McMaster University **Hamilton, ON**
B.Eng. in Computer Engineering *2018 – 2023*

Skills

Languages: Python, C/C++, MATLAB, Java, SQL

ML & Deep Learning: PyTorch, Scikit-Learn, HuggingFace Transformers, Self-Supervised Learning, Transfer Learning, Model Calibration, Knowledge Distillation, Representation Learning, Time-series Modeling

Neuroscience / EEG: MNE-Python, EEG/MEG processing, Clinical EEG/MEG Analysis, Foundation Model Adaptation

Data & Infrastructure: HDF5, HPC (Slurm), Git, CI/CD, Linux

Statistics: GLMs, Hypothesis Testing, Bayesian Models

Publications & Manuscripts

Thesis | 2026

- *Reader-Aware Modeling of Routine Clinical EEG.* Using ordinal report-derived labels that distinguish confident and uncertain interpretations, I examine whether diagnostic certainty is reflected in EEG-derived features, whether ordinal learning captures information lost through binary labeling, and whether residual label variation is associated with the neurologist who authored the report. This thesis investigates routine EEG reports as a scalable source of structured supervision while highlighting the need to account for systematic variation in how clinical labels are generated.

Journal Article | *IEEE Journal of Biomedical and Health Informatics* 2026

- Z. Zhang, G. Medvedev, A. Moiseev, S. Doesburg et al. *Report-derived certainty preserves graded electrophysiological information.* Designed weakly supervised learning algorithms for clinical EEG studies, leveraging LLM-extracted labels and foundation-model representations to learn diagnostic uncertainty from routine clinical report

Conference Paper | *IEEE Sensors Applications Symposium (SAS 2023)* 2023

- Z. Zhang, Y. Shen, J. Valdes, S. Huq, B. Wallace, J. Green, P. Xi, R. Goubran. *Domestic Sound Classification with Deep Learning.* Proposed a lightweight 75K-parameter VGG-like CNN with adaptive pooling and a customized frame-level loss for multi-label sound event detection in domestic environments, achieving performance on par with models 35× larger (CRNN, TCN) while meeting embedded-device (Cortex-M4) compute constraints. Evaluated on the CHiME-Home dataset.

Journal Article | *Am J Physiol Gastrointest Liver Physiol, 325: G295–G305* 2023

- A. Hussain,Z. Zhang, J. Yu, et al. *Haustral Rhythmic Motor Patterns of the Human Large Bowel Revealed by Ultrasound.* Developed image segmentation and frequency analysis software in Python (OpenCV, PyWavelets, scikit-image) to generate spatiotemporal maps of colonic haustral contractions from 5–10 min ultrasound recordings. Identified three distinct rhythmic motor patterns (1, 3, and 12 cpm) in healthy subjects, validating ultrasound as a noninvasive alternative to colonic manometry.

Teaching Experience

Simon Fraser University **Burnaby, BC**
Head Teaching Assistant *2026*

- Led TA teams twice for undergraduate programming, computer architecture, and statistics courses, coordinating labs, grading workflows, and instructional support.
- Developed C++, Python and shell scripts to automate assignment grading and validation, improving consistency and reducing manual grading effort.